ANN_tune

October 31, 2025

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from tensorflow import keras
from keras.models import Sequential
from keras.layers import Dense
from keras_tuner import RandomSearch
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_absolute_error

import keras_tuner as kt
from sklearn.model_selection import KFold
```

```
2025-10-26 18:33:29.895054: I tensorflow/core/util/port.cc:153] oneDNN custom
operations are on. You may see slightly different numerical results due to
floating-point round-off errors from different computation orders. To turn them
off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2025-10-26 18:33:29.895986: I external/local_xla/xla/tsl/cuda/cudart_stub.cc:32]
Could not find cuda drivers on your machine, GPU will not be used.
2025-10-26 18:33:29.898383: I external/local_xla/xla/tsl/cuda/cudart_stub.cc:32]
Could not find cuda drivers on your machine, GPU will not be used.
2025-10-26 18:33:29.905254: E
external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:485] Unable to register
cuFFT factory: Attempting to register factory for plugin cuFFT when one has
already been registered
2025-10-26 18:33:29.916787: E
external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:8454] Unable to register
cuDNN factory: Attempting to register factory for plugin cuDNN when one has
already been registered
2025-10-26 18:33:29.920005: E
external/local_xla/xla/stream_executor/cuda/cuda_blas.cc:1452] Unable to
register cuBLAS factory: Attempting to register factory for plugin cuBLAS when
one has already been registered
2025-10-26 18:33:29.928577: I tensorflow/core/platform/cpu_feature_guard.cc:210]
This TensorFlow binary is optimized to use available CPU instructions in
performance-critical operations.
To enable the following instructions: AVX2 AVX512F AVX512_VNNI AVX512_BF16
```

```
AVX512_FP16 AVX_VNNI AMX_TILE AMX_INT8 AMX_BF16 FMA, in other operations,
rebuild TensorFlow with the appropriate compiler flags.
2025-10-26 18:33:30.425212: W
tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38] TF-TRT Warning: Could not
find TensorRT
```

[4]:
```python
X = pd.read_csv("X_train.csv")
y = pd.read_csv("y_train.csv").squeeze()
X_test = pd.read_csv("X_test.csv")
y_test = pd.read_csv("y_test.csv").squeeze()

print("Train-test data loaded successfully!")
print(f"X_train: {X.shape}, X_test: {X_test.shape}")
```

```
Train-test data loaded successfully!
X_train: (780, 2), X_test: (195, 2)
```

[5]:
```python
X.head()
```

[5]:

	temp	salt
0	24.511438	35.126246
1	22.507472	34.661103
2	25.007761	34.412981
3	24.268401	33.891595
4	25.901783	34.010018

[6]:
```python
sc = StandardScaler()
X = sc.fit_transform(X)
X_test = sc.transform(X_test)
```

[7]:
```python
X.shape
```

[7]: (780, 2)

[9]:
```python
kfold = KFold(n_splits=10)
```

[9]:
```python
from keras.layers import Dropout
```

[10]:
```python
def build_model(hp):
    ann = Sequential()
    for i in range(hp.Int('num_layers',2,20)):
        ann.add(Dense(units=hp.
↪Int('units_'+str(i),min_value=20,max_value=60,step=3),activation='relu',kernel_initializer=
    ann.add(Dense(units=1,activation='linear'))
    ann.compile(optimizer=keras.optimizers.Adam(hp.
↪Choice('learning_rate',[1e-2,1e-3,1e-4])),loss='mean_absolute_error',metrics=[keras.
↪metrics.mae])
```

```
    return ann

tuner = RandomSearch(build_model,objective=kt.Objective('mean_absolute_error',␣
 ↪direction='min'),max_trials=10,executions_per_trial=2,directory='ANN_ABC',project_name='ANN
```

```
/home/kunal/anaconda3/lib/python3.10/site-
packages/keras/src/layers/core/dense.py:87: UserWarning: Do not pass an
`input_shape`/`input_dim` argument to a layer. When using Sequential models,
prefer using an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

[11]:
```
for train_ix, val_ix in kfold.split(X):
    X_train, X_val = X[train_ix], X[val_ix]
    y_train, y_val = y.iloc[train_ix], y.iloc[val_ix]
    tuner.search(X_train,␣
 ↪y_train,epochs=10,validation_data=(X_val,y_val),batch_size=32)
```

```
Trial 10 Complete [00h 00m 08s]
mean_absolute_error: 1.85499769449234

Best mean_absolute_error So Far: 0.8670753538608551
Total elapsed time: 00h 01m 06s
```

[16]:
```
y_train, y_val
```

[16]:
```
(0      6.648872
 1      4.588215
 2      7.028523
 3      5.122176
 4      7.654679
          …
 697    5.801001
 698    4.836807
 699    7.164370
 700    5.385355
 701    6.640552
 Name: chla, Length: 702, dtype: float64,
 702    3.904964
 703    3.417002
 704    7.029361
 705    6.136934
 706    5.501143
          …
 775    5.580679
 776    4.991074
 777    5.076264
 778    3.403436
```

```
779    5.065820
Name: chla, Length: 78, dtype: float64)
```

[17]:
```
tuner.get_best_models()
```

[17]:
```
[<Sequential name=sequential, built=True>]
```

[18]:
```
best_model = tuner.get_best_models(num_models=1)[0]
best_model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 29)	87
dense_1 (Dense)	(None, 50)	1,500
dense_2 (Dense)	(None, 20)	1,020
dense_3 (Dense)	(None, 50)	1,050
dense_4 (Dense)	(None, 47)	2,397
dense_5 (Dense)	(None, 29)	1,392
dense_6 (Dense)	(None, 41)	1,230
dense_7 (Dense)	(None, 53)	2,226
dense_8 (Dense)	(None, 1)	54

Total params: 10,956 (42.80 KB)

Trainable params: 10,956 (42.80 KB)

Non-trainable params: 0 (0.00 B)

[19]:
```
tuner.results_summary()
```

```
Results summary
Results in ANN_ABC/ANN_ABC
Showing 10 best trials
```

```
Objective(name="mean_absolute_error", direction="min")

Trial 07 summary
Hyperparameters:
num_layers: 8
units_0: 29
units_1: 50
learning_rate: 0.001
units_2: 20
units_3: 50
units_4: 47
units_5: 29
units_6: 41
units_7: 53
units_8: 44
units_9: 38
units_10: 38
units_11: 44
units_12: 32
units_13: 47
units_14: 44
units_15: 59
units_16: 29
units_17: 38
units_18: 53
Score: 0.8670753538608551

Trial 02 summary
Hyperparameters:
num_layers: 17
units_0: 56
units_1: 32
learning_rate: 0.01
units_2: 50
units_3: 53
units_4: 35
units_5: 26
units_6: 56
units_7: 20
units_8: 20
units_9: 20
units_10: 20
units_11: 20
units_12: 20
units_13: 20
units_14: 20
units_15: 20
units_16: 20
```

```
Score: 0.8678410351276398

Trial 05 summary
Hyperparameters:
num_layers: 11
units_0: 53
units_1: 38
learning_rate: 0.001
units_2: 56
units_3: 35
units_4: 59
units_5: 23
units_6: 32
units_7: 53
units_8: 41
units_9: 26
units_10: 47
units_11: 56
units_12: 35
units_13: 23
units_14: 59
units_15: 23
units_16: 44
Score: 0.8715607523918152

Trial 00 summary
Hyperparameters:
num_layers: 5
units_0: 26
units_1: 23
learning_rate: 0.001
units_2: 20
units_3: 20
units_4: 20
Score: 1.2680396735668182

Trial 06 summary
Hyperparameters:
num_layers: 19
units_0: 41
units_1: 50
learning_rate: 0.0001
units_2: 50
units_3: 20
units_4: 44
units_5: 56
units_6: 53
units_7: 20
```

```
units_8: 47
units_9: 38
units_10: 29
units_11: 50
units_12: 23
units_13: 41
units_14: 47
units_15: 38
units_16: 59
units_17: 20
units_18: 20
Score: 1.5274211168289185

Trial 03 summary
Hyperparameters:
num_layers: 15
units_0: 41
units_1: 59
learning_rate: 0.0001
units_2: 23
units_3: 50
units_4: 53
units_5: 26
units_6: 53
units_7: 59
units_8: 20
units_9: 20
units_10: 20
units_11: 47
units_12: 38
units_13: 56
units_14: 56
units_15: 50
units_16: 47
Score: 1.6488606333732605

Trial 09 summary
Hyperparameters:
num_layers: 15
units_0: 29
units_1: 50
learning_rate: 0.0001
units_2: 50
units_3: 56
units_4: 35
units_5: 50
units_6: 23
units_7: 23
```

```
units_8: 38
units_9: 50
units_10: 59
units_11: 26
units_12: 20
units_13: 35
units_14: 41
units_15: 26
units_16: 29
units_17: 23
units_18: 29
Score: 1.85499769449234

Trial 04 summary
Hyperparameters:
num_layers: 13
units_0: 44
units_1: 53
learning_rate: 0.0001
units_2: 53
units_3: 20
units_4: 50
units_5: 47
units_6: 32
units_7: 26
units_8: 53
units_9: 20
units_10: 53
units_11: 47
units_12: 26
units_13: 32
units_14: 20
units_15: 20
units_16: 32
Score: 1.972251534461975

Trial 08 summary
Hyperparameters:
num_layers: 12
units_0: 59
units_1: 29
learning_rate: 0.0001
units_2: 53
units_3: 56
units_4: 32
units_5: 29
units_6: 47
units_7: 32
```

```
units_8: 41
units_9: 29
units_10: 32
units_11: 26
units_12: 47
units_13: 20
units_14: 26
units_15: 53
units_16: 23
units_17: 29
units_18: 26
Score: 1.9942893385887146

Trial 01 summary
Hyperparameters:
num_layers: 7
units_0: 59
units_1: 29
learning_rate: 0.0001
units_2: 32
units_3: 56
units_4: 26
units_5: 20
units_6: 20
Score: 2.103670120239258
```

[ ]:

```

```

[ ]:

```

```

[ ]:

```

```